Lab 2: Tableau

Spring COMP4462

Before the lab

- Set up Tableau
- Instruction on Canvas assignment and announcement

1. Sign into an existing Tableau.com account, or create a new account using your school-issued email
2. Once signed in, visit the Academic Quick Start page to download the latest versions of Tableau Desktop and Tableau Prep Builder
3. Activate with product key: [redacted] Note that this product key is exclusively for students of this course; there will be strong repercussions if I find someone else has used it to activate Tableau.
4. Already have a copy of Tableau Desktop installed? Update the license key in the application: Help menu → Manage Product Keys

Having trouble? The Academic Quick Start page includes FAQs and help articles.

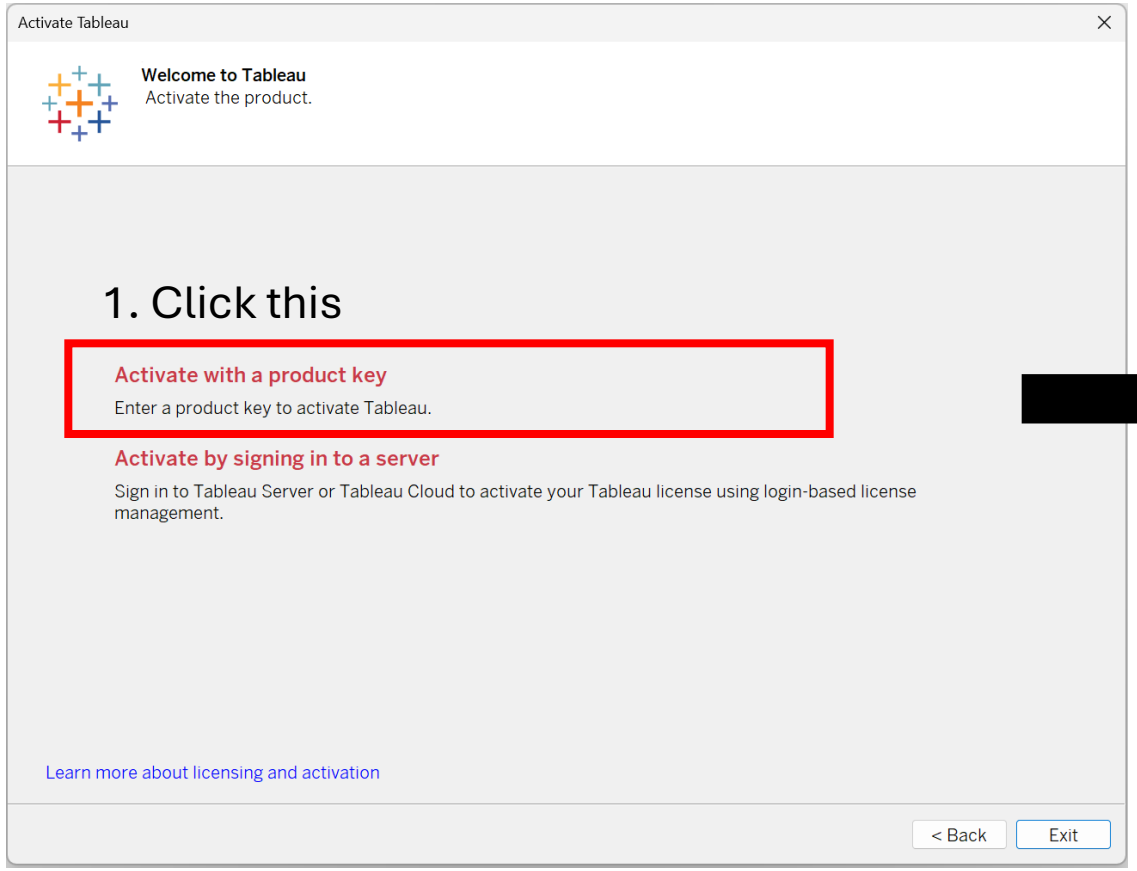
Some helpful resources:

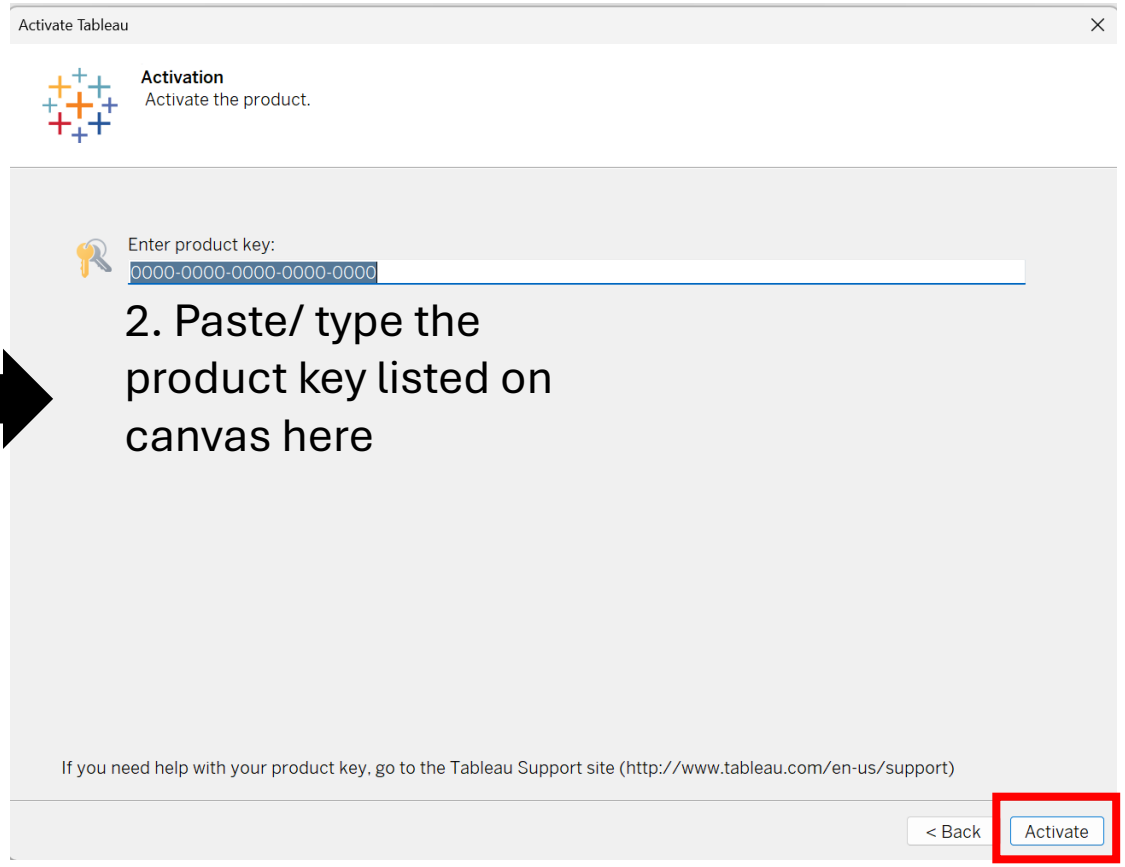
- Check out Tableau's Free Training Videos and Tableau Public's How-To Videos
- Join the Tableau Community and explore the Forums to search and ask questions
- For product support, review the Knowledge Base and Tableau Help platform

1. Once setup, we recommend that you familiarize yourself with the visualization techniques and the user interface of Tableau. Watch tutorials about the system and read articles that explain and describe it. We have uploaded to Canvas a tutorial video on how to use the system -- It is **highly** recommended that you watch the video. It is 20 minutes long and showcases many, many capabilities of the system.
2. Also, there is extensive documentation online. The lecture slides from the Tableau day show more of this material in detail. You will definitely need to spend some time on your own consulting this online documentation, the available books, etc., in order to learn the system deeply.

Please refer to the product key on Canvas

Set up tableau

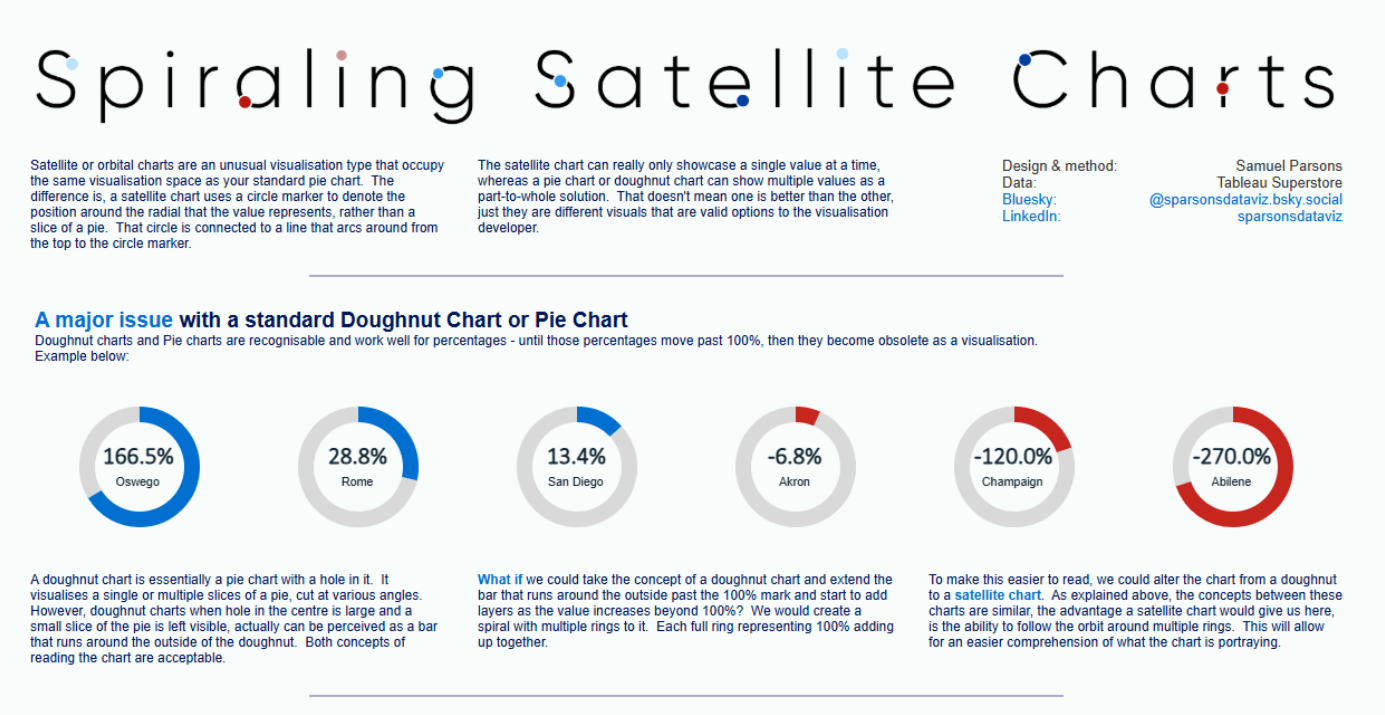




3. Click this

What is Tableau?

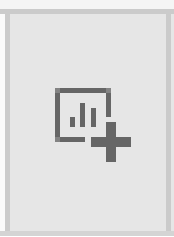
- State of the art data visualization software
- Creates interactive dashboards and data stories
- Open community to share your viz to public



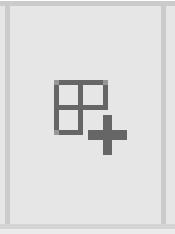
Credits to: Samuel Parsons
url:
https://public.tableau.com/app/profile/sparsonsdataviz/viz/SpiralingSatelliteCharts/SpiralingSatelliteCharts

3 views of Tableau

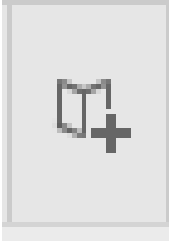
- Worksheet
  - Where you make the specific charts and analyses data



- Dashboard
  - Where you combine worksheets
  - Interactivity



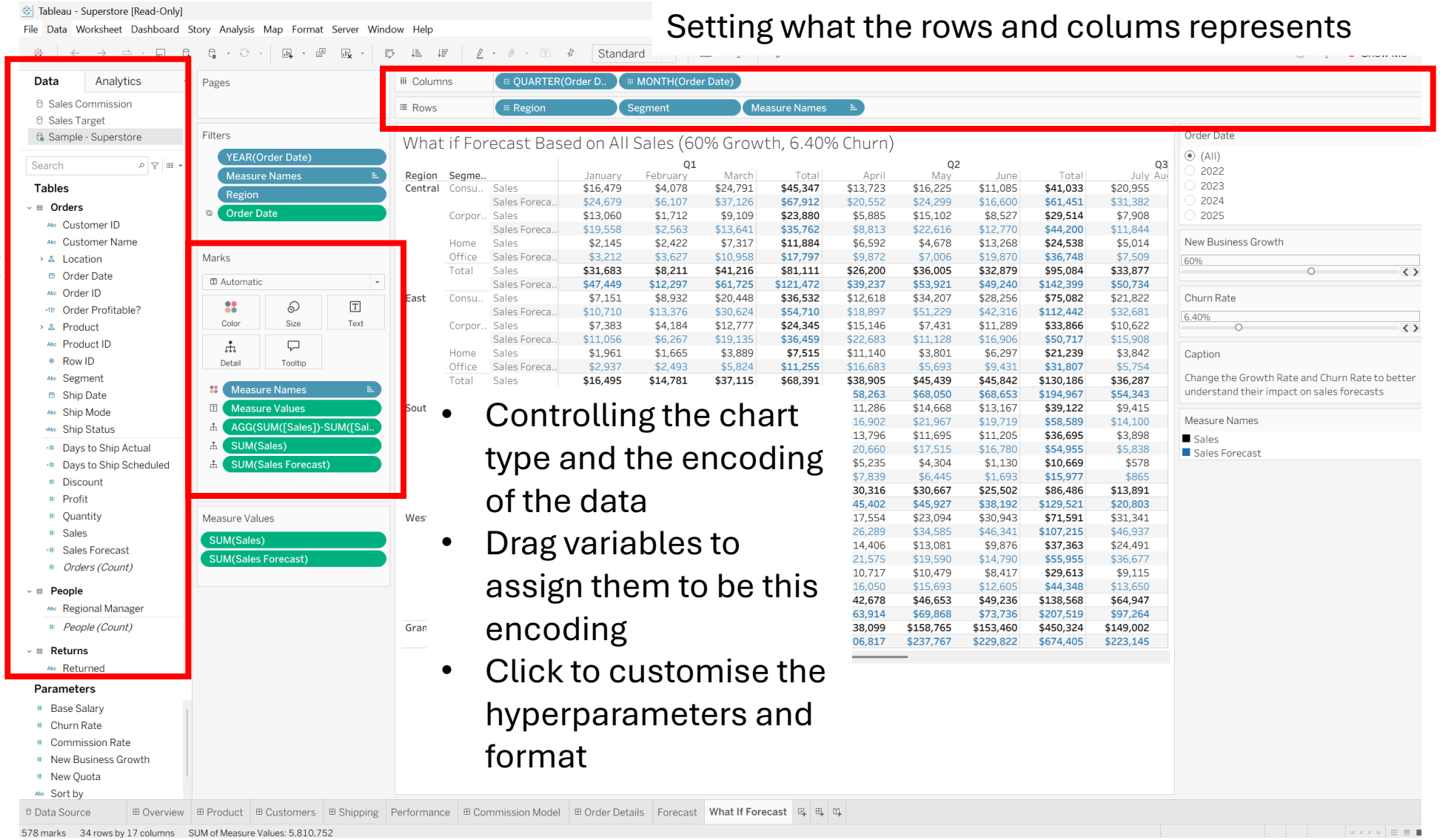
- Story
  - Like a storybook, adds more descriptive text and order
  - Combines dashboards



Worksheet

Choosing variables/ accessing variables

Setting what the rows and colums represents



- Controlling the chart type and the encoding of the data
- Drag variables to assign them to be this encoding
- Click to customise the hyperparameters and format

Difference in variables

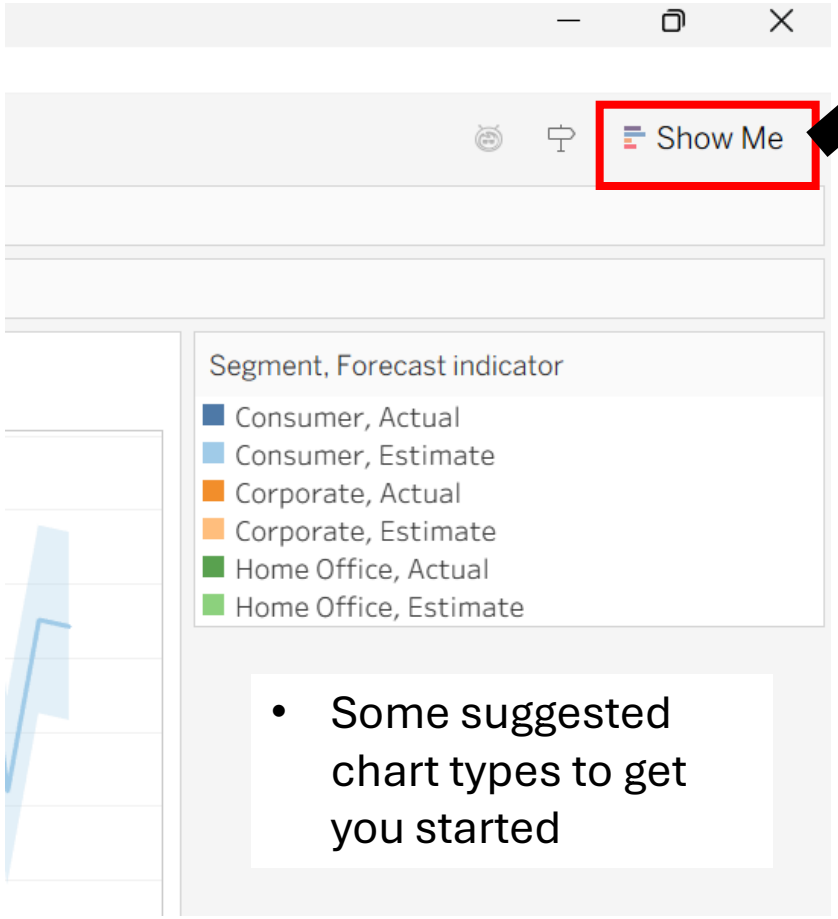
- Dimensions (ABC)
  - Qualitative
  - Discrete
- Measures (#)
  - Quantitative
  - Continuous

Tables

- Abc Beverage
- Abc Beverage category
- Abc Beverage prep
- Abc Caffeine (mg)
- Abc *Measure Names*

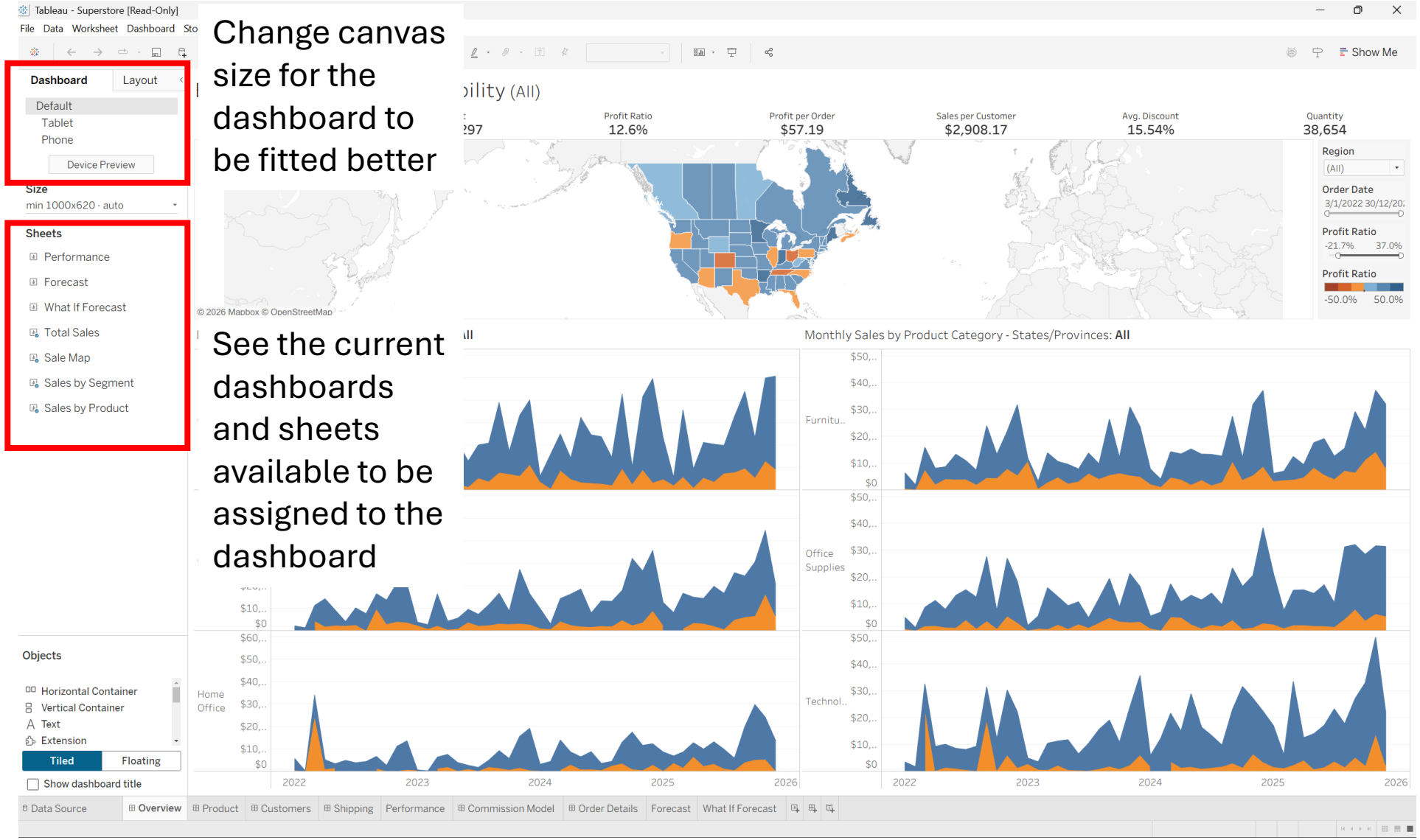
- # Calories
- # Cholesterol (mg)
- # Dietary Fibre (g)
- # Protein (g)
- # Sodium (mg)
- # Sugars (g)
- # Total Carbohydrates (g)
- # Total Fat (g)
- # *starbucksdrinks (Count)*
- # *Measure Values*

Fun functions for the clueless



- Some suggested chart types to get you started

Dashboard

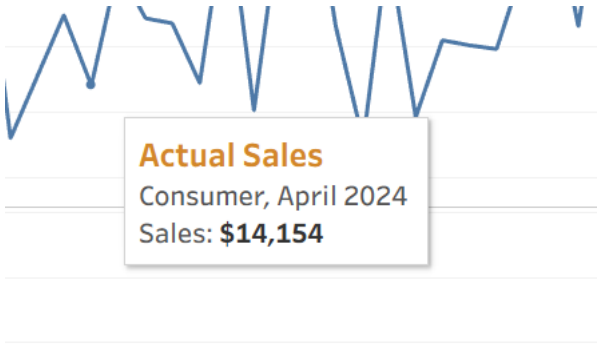


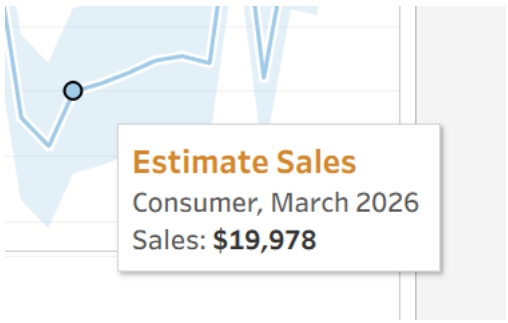
Change canvas size for the dashboard to be fitted better

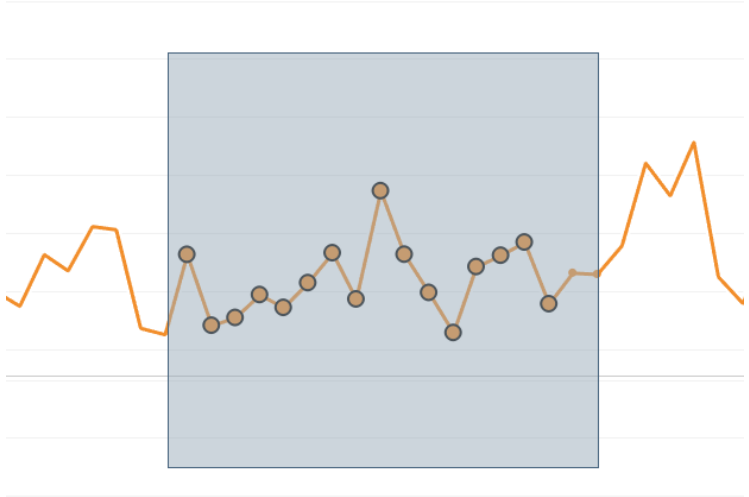
See the current dashboards and sheets available to be assigned to the dashboard

Interactions

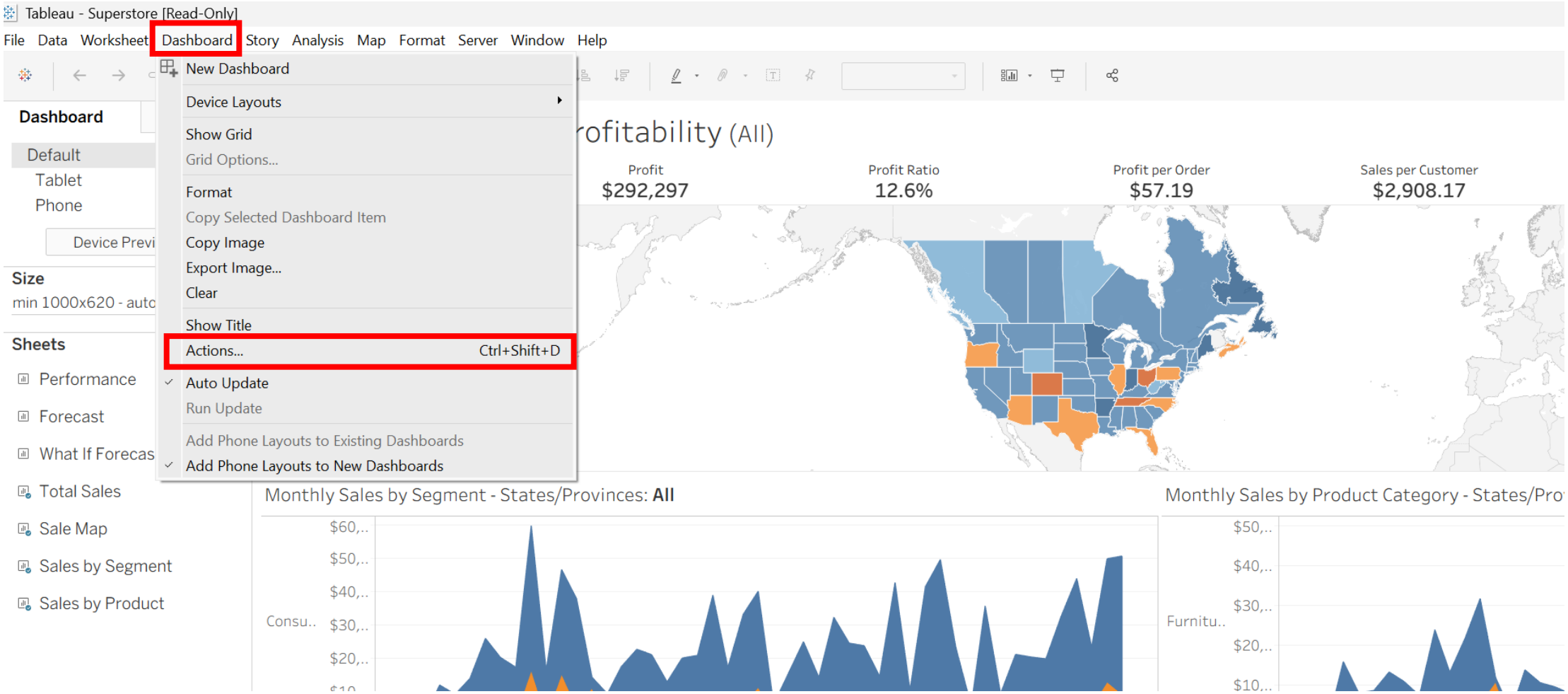
- Tableau allows interactions
- Can be interactive single sheet or cross referencing sheets
- Interactions include: hover, click, brushing…







To introduce interactivity



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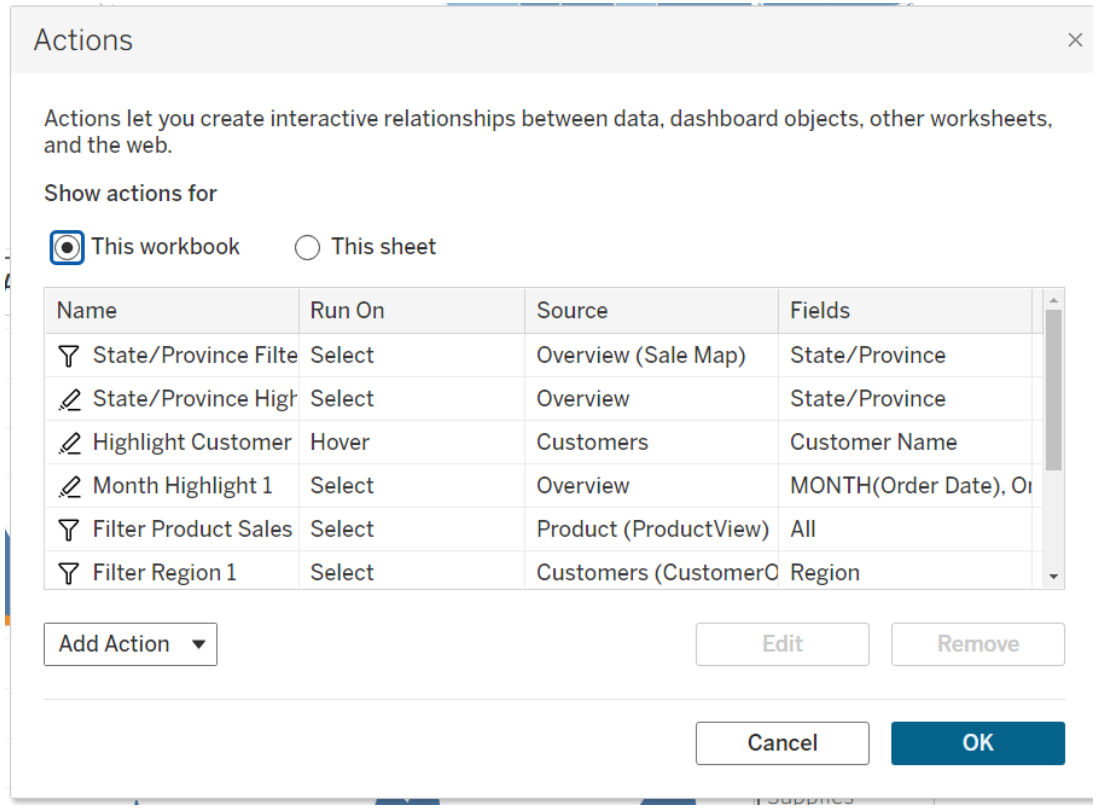
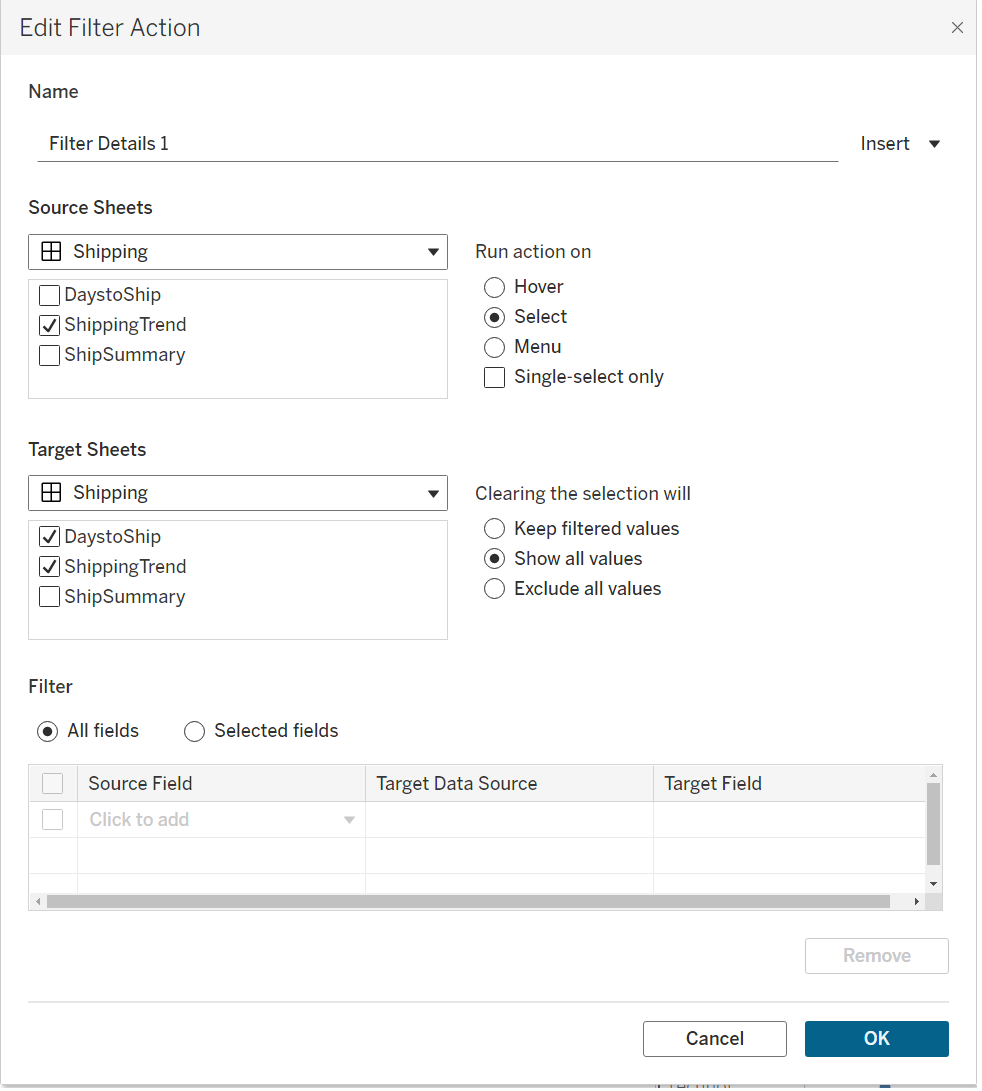
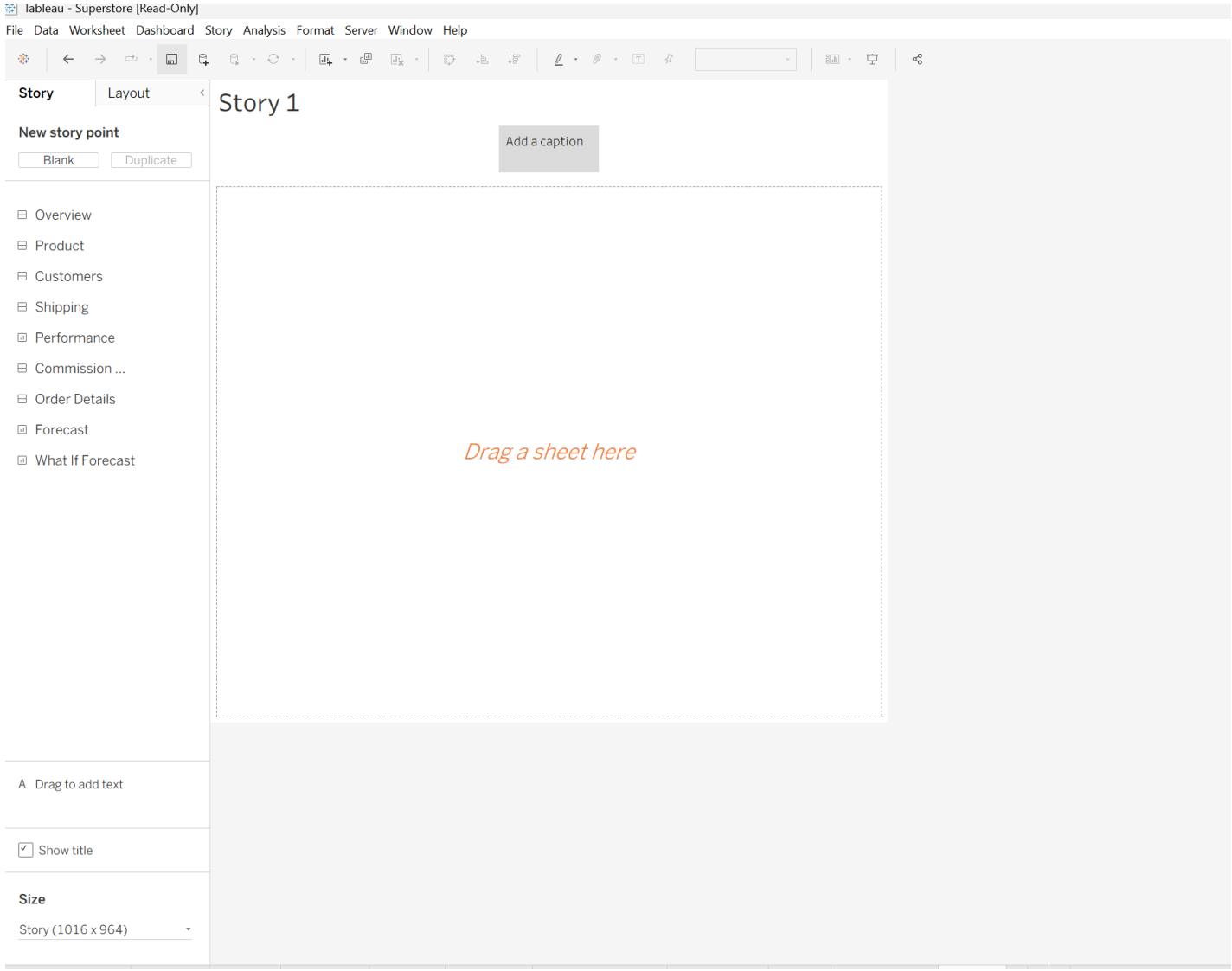




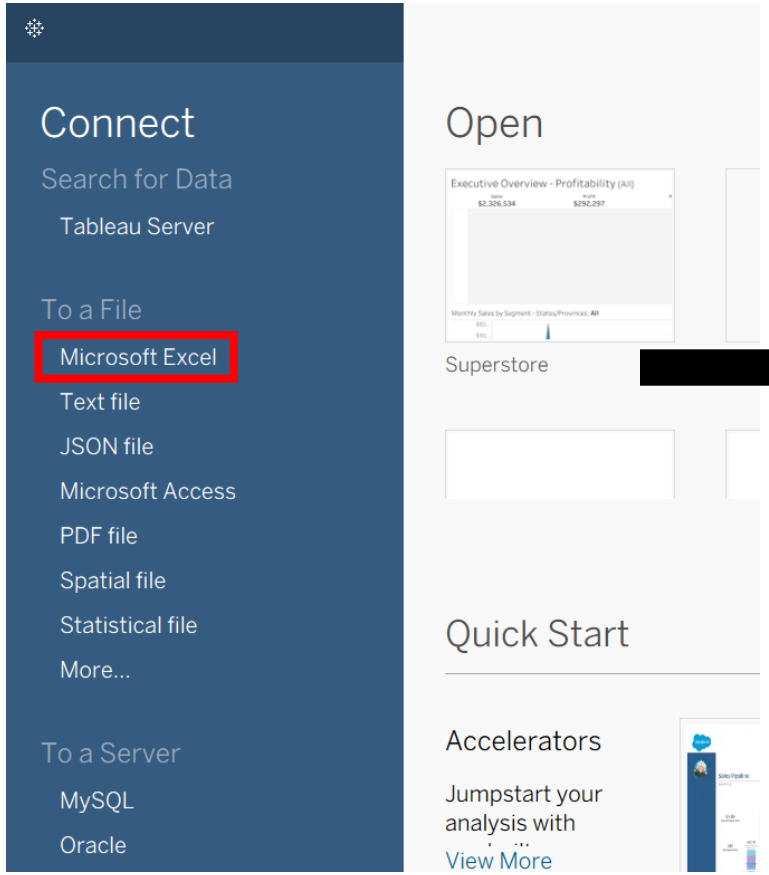
StoryPoints

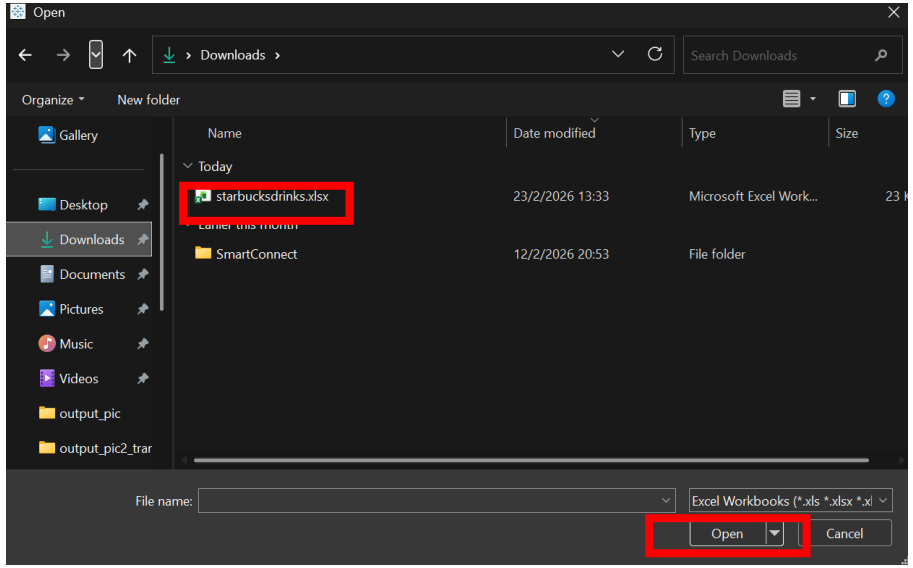
- Like a ppt slide to better organise all the dashboards to form the presentation



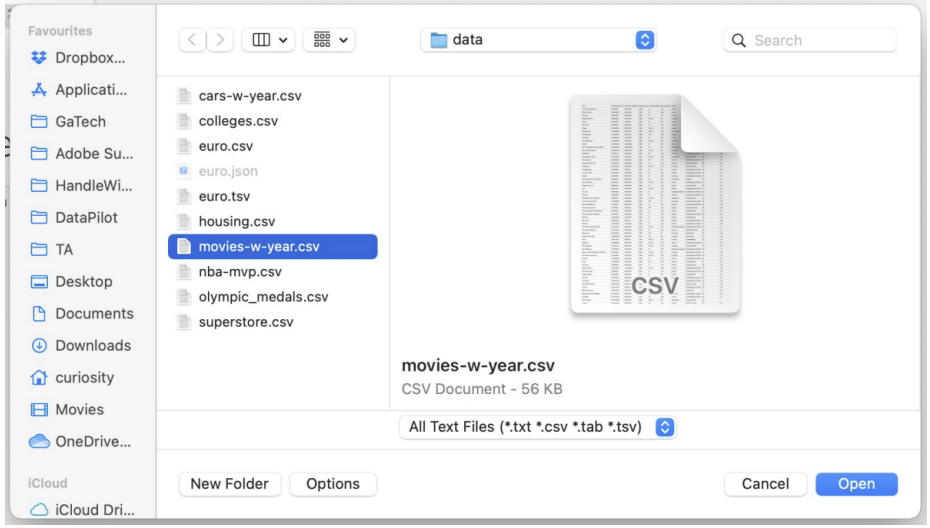
Lets try out tableau!

- Step 1: input data





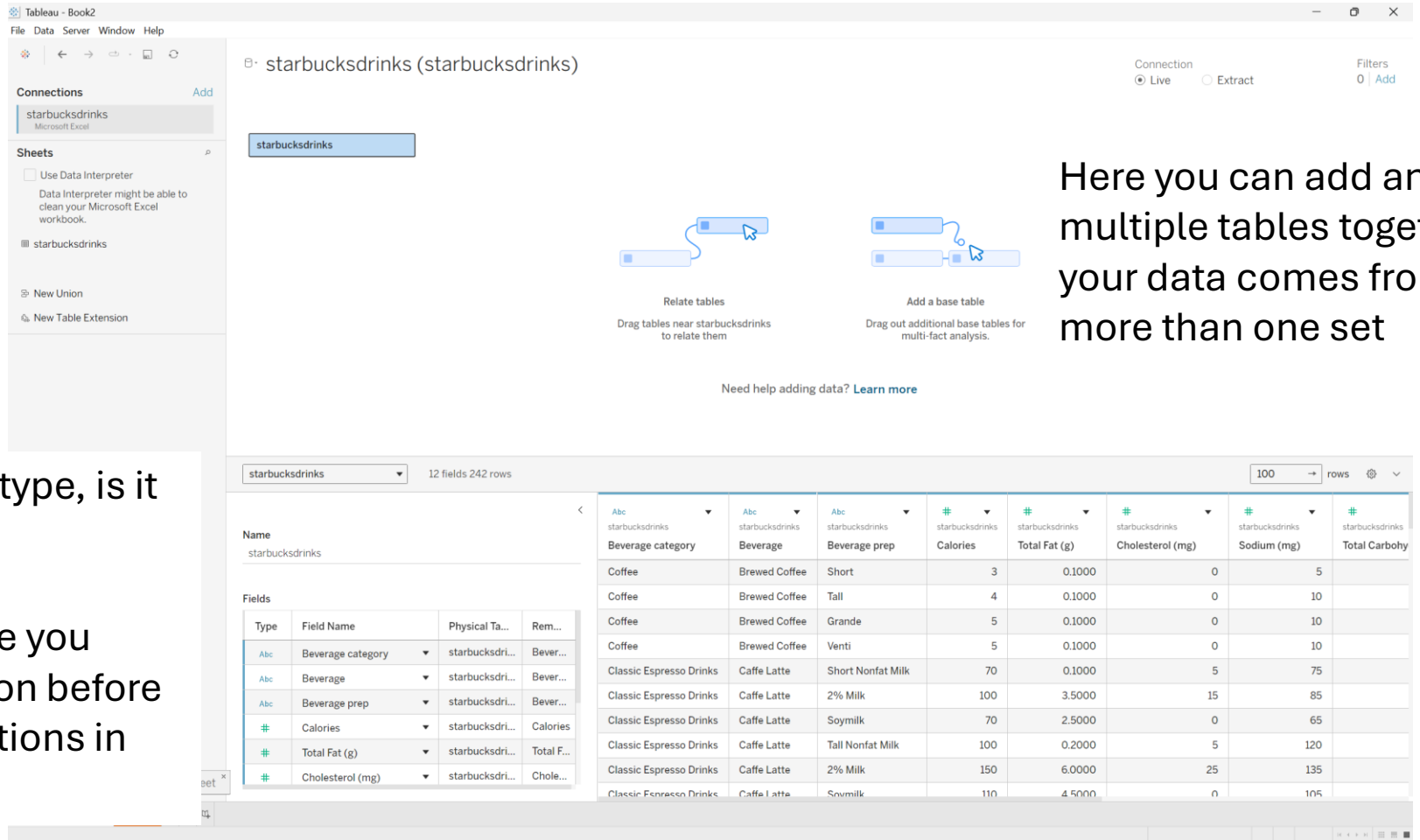
For Windows users



For Mac users

Lets try out tableau!

- Step 2: Check data



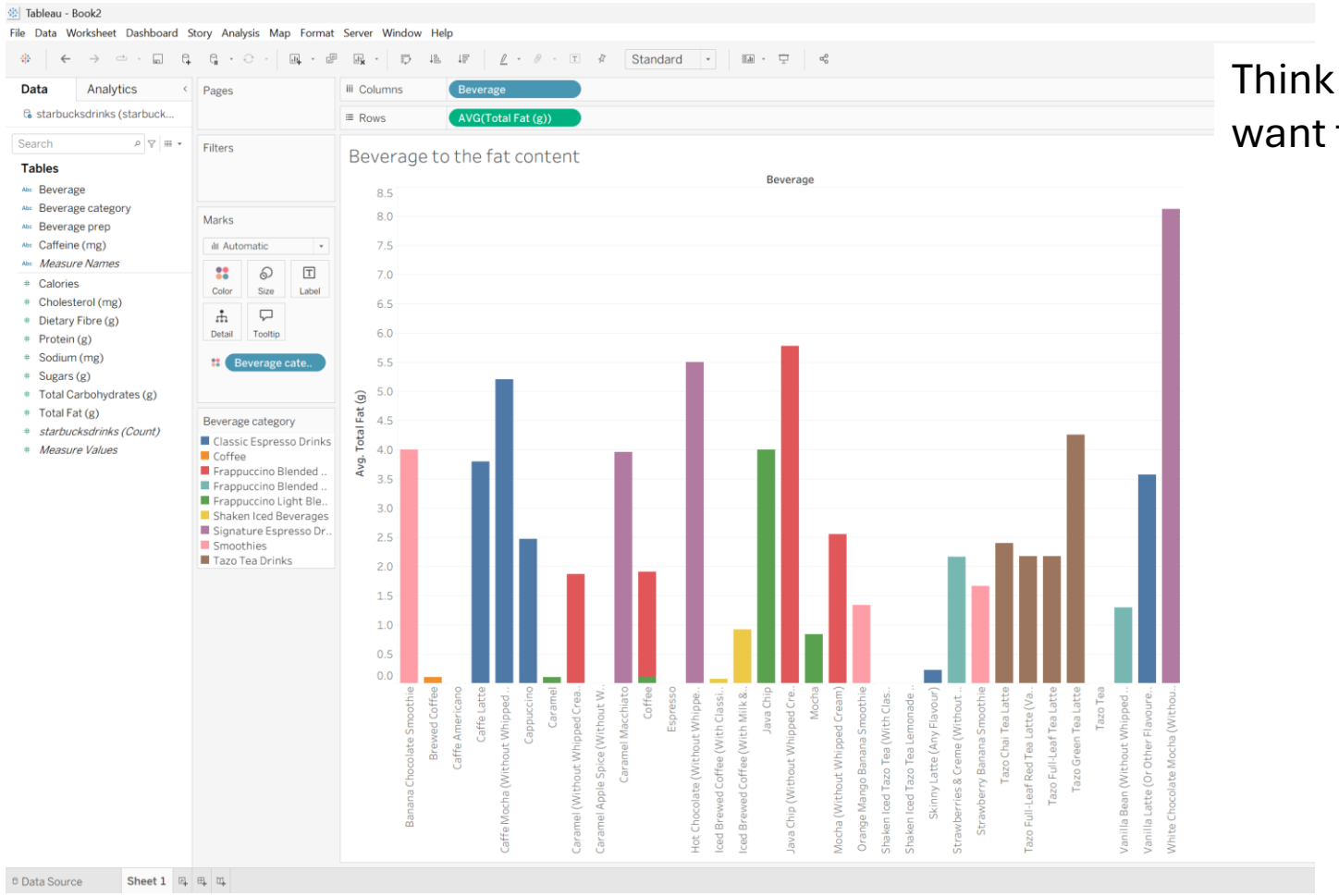
Here you can add and join multiple tables together if your data comes from more than one set

Check the data type, is it reasonable?

Also recall: Have you seen this function before in other applications in Lab 1?

Lets try out tableau!

- Step 3: Plot out data in worksheet



Think about the task you want to achieve

Problems you might encounter

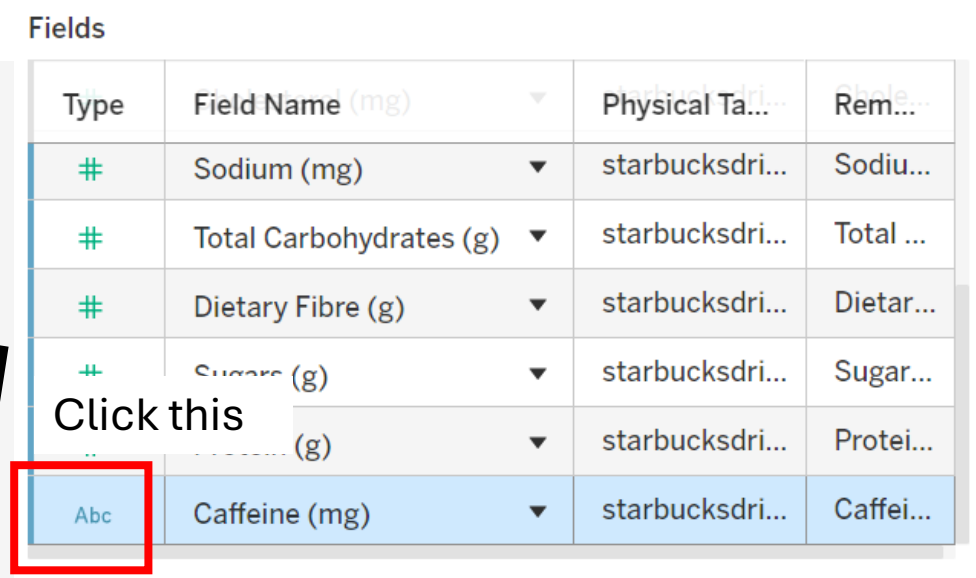
- Wrong variable type
  - E.g. Caffeine is a dimension, but logically it should be a measure
- Making derived fields
  - E.g. modelling a ranking metric based on an equation using fat, sugar, and calories
- The chart types are boooooooring
  - I want to use a sunburst chart!

HOW TO SOLVE THESE?

Wrong data type?

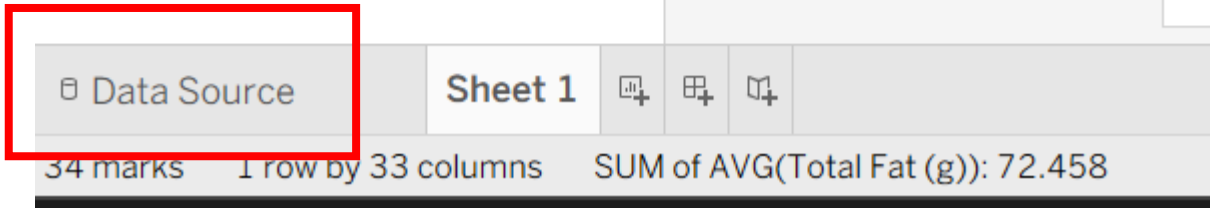
- Solution: go back to data view and check!

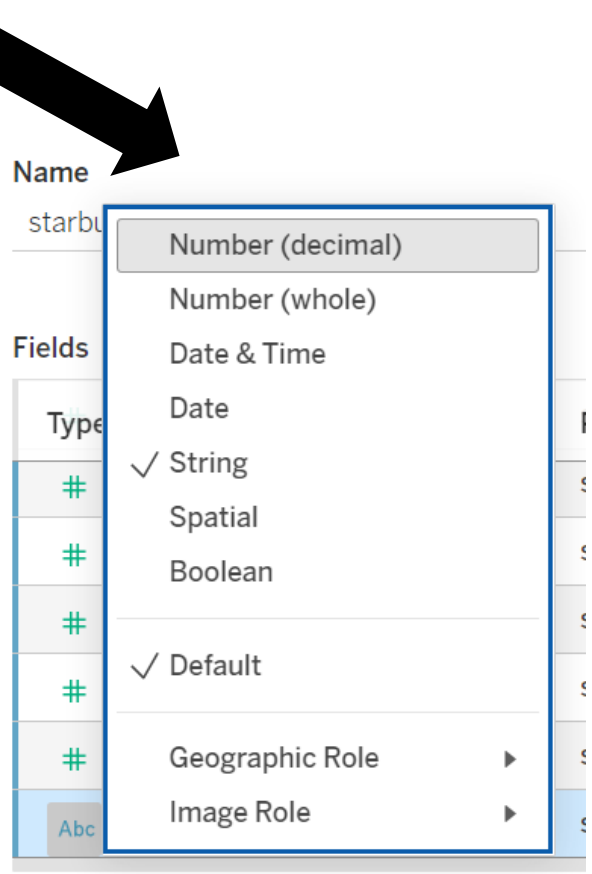
REMEMBER:
Check the data to check if it is really all numbers



Click this

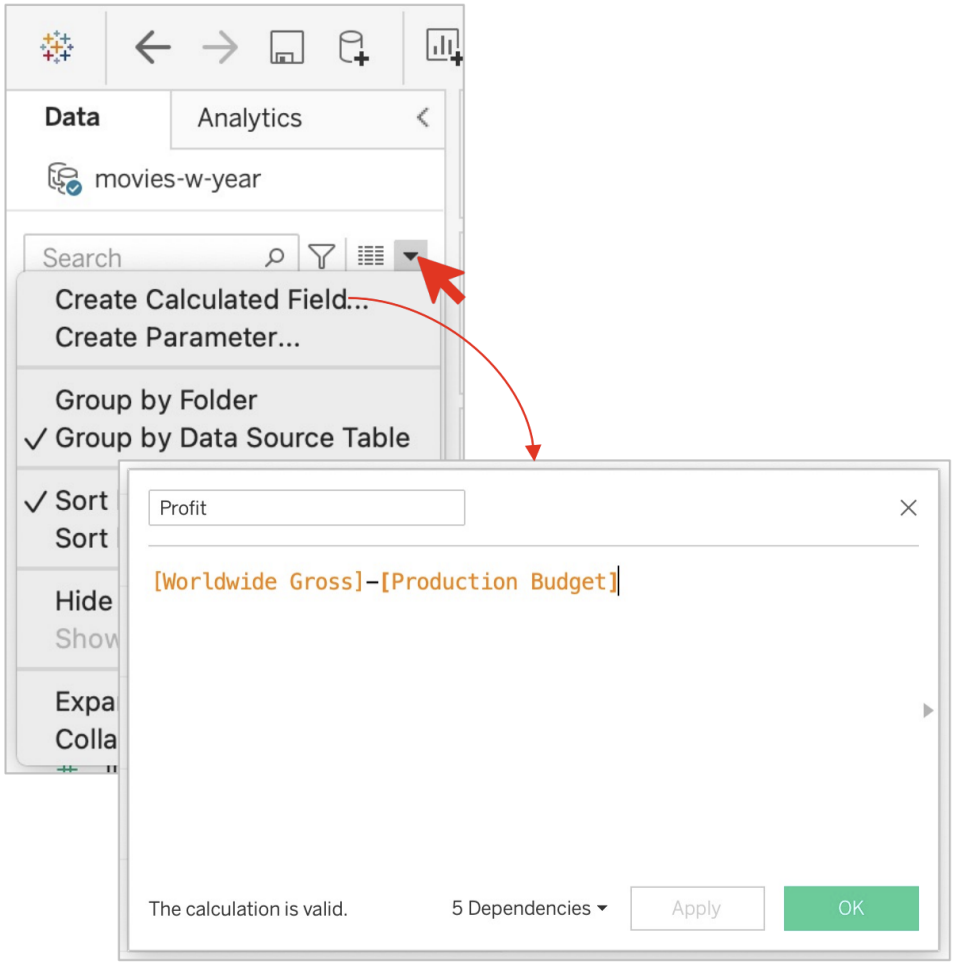
At the bottom left corner





Calculated fields

- Custom fields to derived required data
- Or customise parameters to better control the filters and other variables



Want more charts?

- Use extensions!
  - Open community that have many different available chart types
  - Remember to check usage terms (some might have restriction as to if it can be used for commercial use)
- Online tutorials
  - Some charts, like parallel coordinate/ stacked barchart with the split in the center can be created by some tinkering
  - Good source is to find how to do so online, or toying around yourself!

More about Tableau

- So much more on tableau to explore!
- The video posted on canvas should cover the basics
- Explore other's dashboards on Tableau Public!

Lab Task

- On Canvas, under https://canvas.ust.hk/courses/67750/assignments/412329

Reminder

- Lab task deadline: Feb 27 23:59
- Take attendance if you have not yet!
- Group project GP-2 deadline!

Questions?